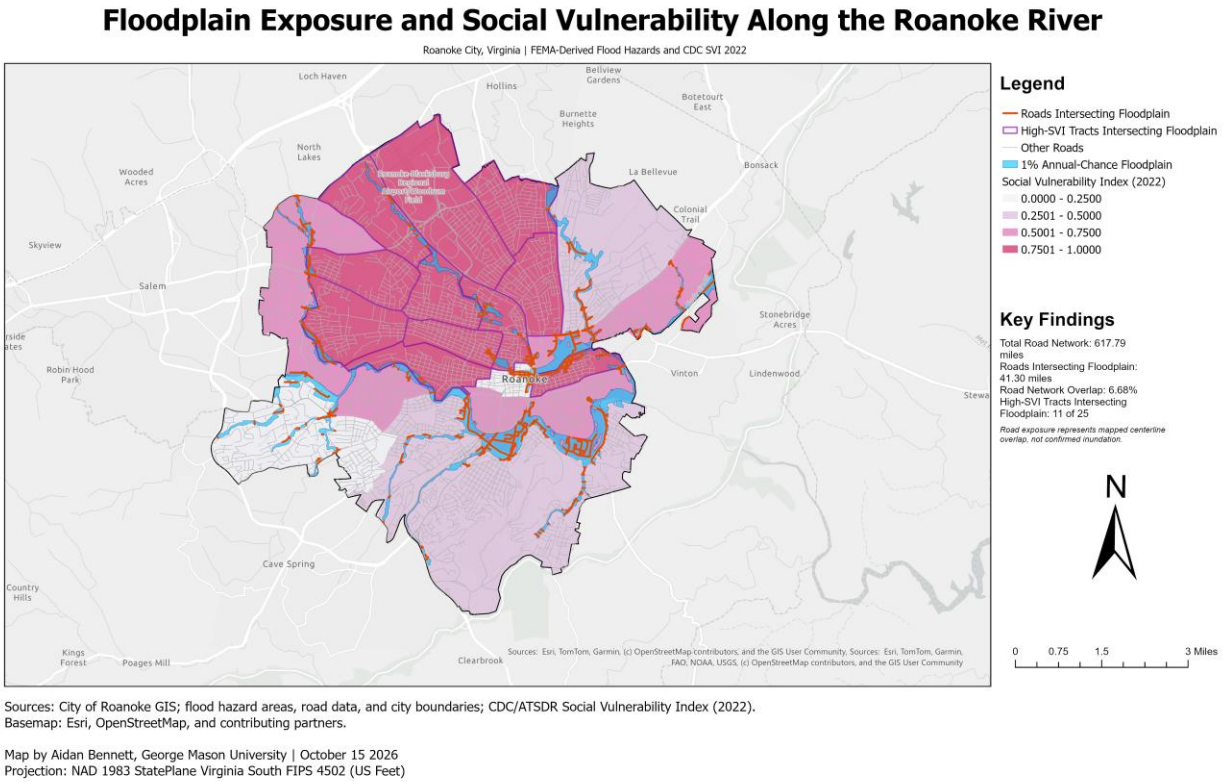
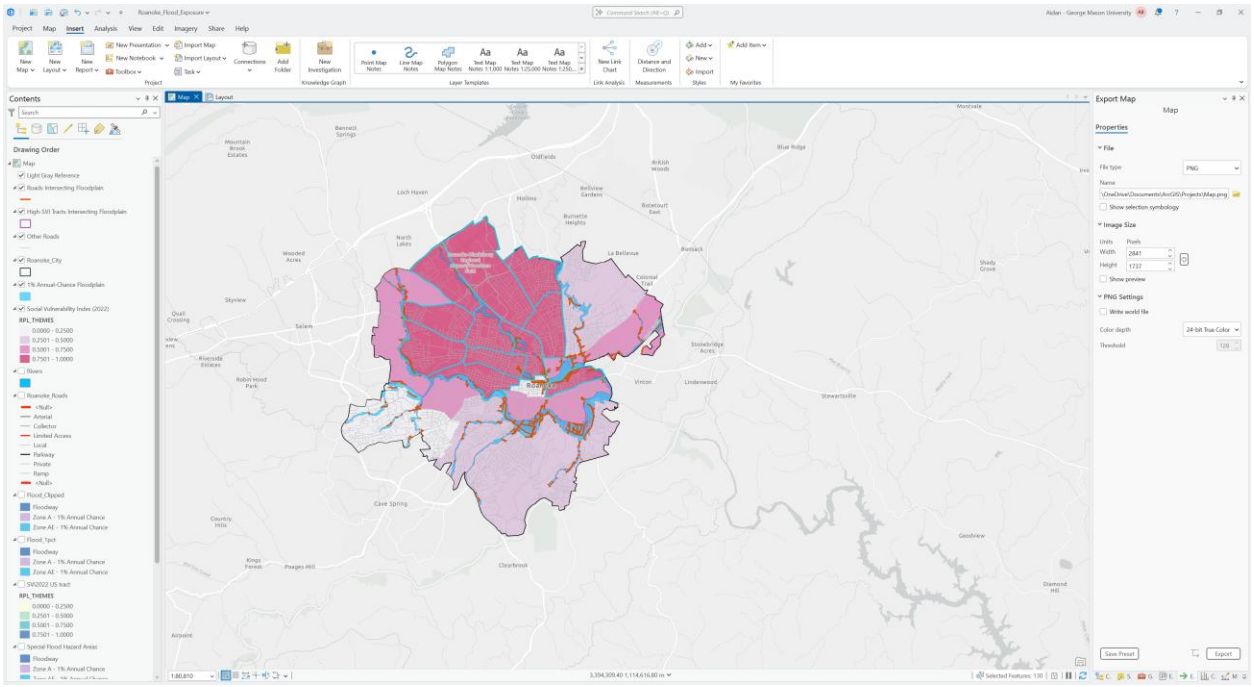
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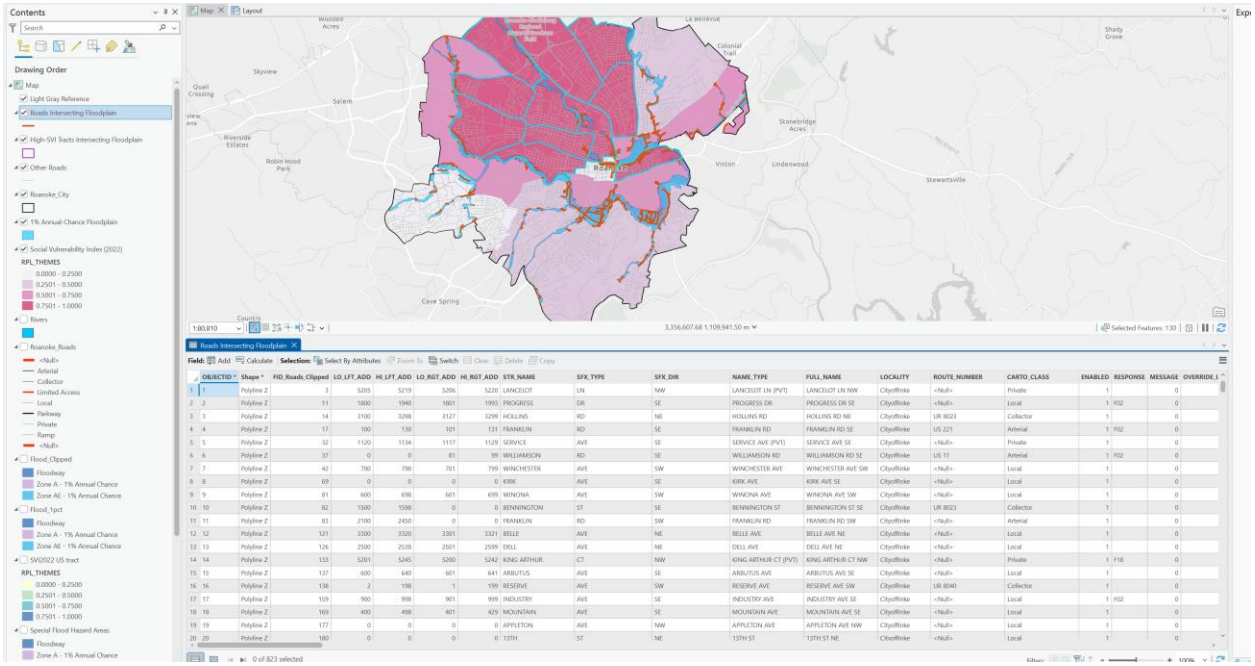
Sources: City of Roanoke GIS; flood hazard areas, road data, and city boundaries; CDC/ATSDR Social Vulnerability Index (2022).
Basemap: Esri, OpenStreetMap, and contributing partners.

Map by Aidan Bennett, George Mason University | October 15 2026
Projection: NAD 1983 StatePlane Virginia South FIPS 4502 (US Feet)

Screenshot of overall map and contents:



Attribute table example:



Floodplain Exposure and Social Vulnerability Along the Roanoke River

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Flooding is a major environmental issue, since its effects go beyond water itself. Floodwater can destroy transportation, isolate neighborhoods, and damage property, which complicates emergency responses. Communities do not necessarily experience these disruptions equally. Eisenhauer and Finley (2023) explain that economic and social disadvantages can shape unequal flood impacts and recovery outcomes. This project examines these overlapping concerns in Roanoke City, Virginia, using geographic information systems (GIS). The overall question I put together is: Which portions of the city's road network connect with mapped flood hazards, and where do floodplain areas coincide with census tracts showing relatively high social vulnerability, and why?

Roanoke City is in southwestern Virginia. It is where the Roanoke River and associated waterways pass through developed areas. The city is a useful study area because the streets, neighborhoods, and river corridors are under the same urban landscape. For the physical hazard layer, I used flood-zone polygons from the City of Roanoke's data. The analysis focused on the 1%-annual-chance floodplain, commonly called the 100-year floodplain. This means the modeled event has a 1% chance of being equaled or exceeded in any year and does not mean flooding occurs only once every century (Federal Emergency Management Agency [FEMA], n.d.).

I also found some city road centerlines data and an administrative boundary from municipal GIS services (City of Roanoke, n.d.-a, n.d.-b, n.d.-c). For the social component, I used the 2022 Social Vulnerability Index (SVI) census tract data published by the Centers for Disease Control and Prevention and Agency for Toxic Substances and Disease Registry (ATSDR, n.d.). The overall ranking field, RPL_THEMES, measures relative vulnerability using sixteen social variables grouped into four themes (ATSDR, 2024).

I created the project and used NAD 1983 StatePlane Virginia South FIPS 4502 coordinate system for it. I first went and exported the road network as well as the city boundary to the local file geodatabase to organize my work. I used the clip tool to be able to restrict the roads for Roanoke City. Then, I used the select by attributes tool to retain floodway and Zones A and AE. This represents the chosen 1% annual chance flood hazard categories, while not showing the 0.2% annual chance for Zone X. After exporting these, I clipped them to the city boundary and then used the dissolve tool to create one layer for the analysis for the floodplain.

I then used the intersect tool to be able to create new road line geometries only where the roads and floodplain polygons overlap, which I used on the dissolved floodplain and the clipped road centerlines. Using the calculate geometry attributes tool, I found their lengths in statute miles, and I also calculated the total length of all clipped city road centerlines. This allowed the overlapping share of the network to be appraised. This allowed me to measure horizontal geographic overlap, rather than just observed road closures.

For social vulnerability, I filtered the national SVI layer using Roanoke City's county-equivalent FIPS code, which is 51770. I then exported 25 census tracts and then classified the SVI percentile field into 4 rangers, from 0 to 1. I had to define the highest national vulnerability quartile, which is set to RPL_THEMES at at least 0.75. After that, I used a spatial subset selection to find which of the higher-SVI tracts intersected with the floodplain, then exported them.

I marked my map to use purple and pink tract shading for these results, in addition to blue for floodplain polygons, orange road lines, and outlined higher-SVI tracts. (With this, a simple legend, north arrow, and scale bar were also added with the projection and credits, and the summary as well.)

The analysis that I conducted on the roads showed 617.79 miles of city road centerlines, and of those 617 miles, 41.30 miles came across the floodplain. The other 576.49 miles were outside of the polygons, leaving 93.32% outside. Figure 1 compares the total road centerline mileage both inside and outside of the floodplain. Figure 2 shows the percentage split between the exposed and unexposed mileage. The orange road segments are not evenly distributed.

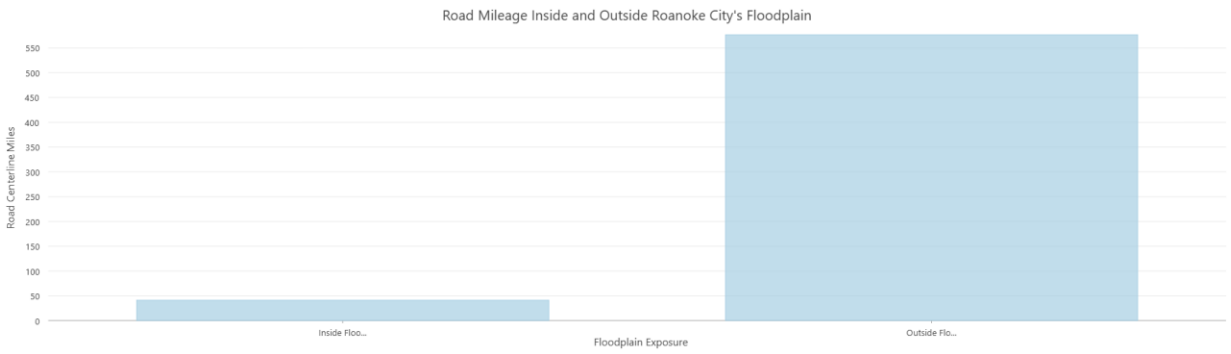


Figure 1: Road Centerline Mileage Inside and Outside the 1%-Annual-Chance Floodplain in Roanoke City

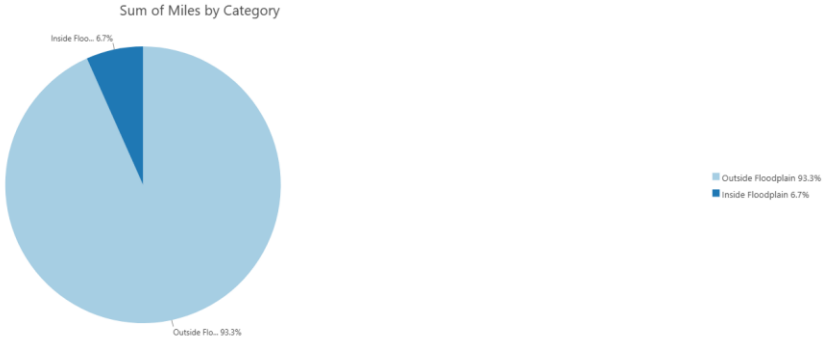


Figure 2: Percentage of Roanoke City's Road Network Inside and Outside the 1%-Annual-Chance Floodplain

The analysis that selected 11 of the 25 census tracts from Roanoke city met two of my criteria for the project: an overall national SVI ranking of at least 0.75, as well as some mapped floodplain overlap. The higher ranked tracts are shown with darker colors, and the tracts indicate where social vulnerability relatively coincides with the floodplain geography.

Something important that I discovered is not just that flooding may affect the infrastructure, but also that some potentially exposed spots are in or near communities that may struggle with proper preparation, response, and recovery in the event of an emergency. Road networks can serve as transportation routes and are important for emergency access and evacuation. That being said, local disruption can also require different routes. The map shows the importance of social vulnerability as well as transportation exposure should be considered both at the same time during environmental planning.

The analysis shows an initial screening exercise and does not show a forecast for any particular storm. Flood zones produced by FEMA show modeled probabilities where rainfall, stream conditions, and local drainage could potentially create different flood patterns. A road may intersect with a polygon, even if a bridge is above floodwater. This can also be said with damaging urban flooding, which can occur outside of the mapped 1%-annual-chance floodplain.

The calculation also represents road centerlines, not roadway pavement area, traffic volumes, emergency accessibility, or the number of people affected. Likewise, a census tract with any floodplain intersection was counted even if the overlap was small. The SVI ranking describes a tract, not every household within it, and the project did not calculate how many

residents actually live inside the hazard areas. Differences in dataset dates and spatial accuracy introduce additional uncertainty.

This project helped map flood hazards as well as transportation infrastructure and census tract vulnerability, which provides a better picture of potential environmental exposure to Roanoke City. The results are clear with it identifying 41.30 miles of road centerlines within the mapped floodplain, as well as 11 high-SVI tracts with some overlap.

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